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# The evolution of genetic topologies

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**Abstract**

This manuscript explores the simultaneous evolution of population genetic parameters and topological features within a population graph through a series of Monte Carlo simulations. I show that node centrality and graph breadth are significantly correlated to population genetic parameters $\Phi_{ST}$ and $\hat{M}$ ($\rho = -0.95$; $\rho = -0.98$, respectively), which are commonly used in quantifying among population genetic structure and isolation by distance. Next, the topological consequences of migration patterns are examined by contrasting $N$-island and stepping stone models of gene movement. Finally, I show how variation in migration rate influences the rate of formation of specific topological features with particular emphasis to the phase transition that occurs when populations begin to become fixed due to restricted movement of genes among populations. I close by discussing the utility of this method for the analysis of intraspecific genetic variation.
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## 1. Introduction

The observed distribution of genetic variation within species is an amalgamation of both historical and ongoing evolutionary and demographic processes. The manner by which these interacting processes have in the past and continue to shape genetic variation remains a key goal of evolutionary biology. In this paper, I examine the evolution of intraspecific genetic structure within the population graph framework described in Dyer and Nason (2004) with special emphasis on the effects gene movement have on the evolution of genetic structure.

A population graph, $G$, contains two sets ($G = \{V, E\}$), where the node set, $V$, represents a finite number of populations (of size $|V| = K$) within which genetic information has been ascertained from their constituent individuals. While Dyer and Nason (2004) introduced the population graph framework using co-dominant genetic markers, other genetic information is equally as amenable to this approach. The edge set is a two-element subset of $V$ (e.g., $E \subseteq [V]^2$) which defines the pairs of populations whose genetic structures are conditionally dependent. That is to say, populations $v_i$ and $v_j$ will share an edge if and only if there is significant genetic covariance between the populations *after* removing the covariation each population has with all the remaining populations in the data set. Conceptually, conditional independence is similar to the Type III sums of squares, or sums of squares to add, in a linear model in that the amount of variation a candidate variable explains, conditional on all other variables already in the model, can be ascertained (Whittaker, 1990). For the purposes of this manuscript, I define conditional independence of populations $v_i$ and $v_j$, given the remaining populations, say $v_k$; $k \neq i \neq j$, as $v_i \perp\!\!\!\perp v_j | v_k \equiv P(v_i \cap v_j | v_k) = P(v_i|v_k)P(v_j|v_k)$. This definition is central to the *topology*, or pattern of population $\leftrightarrow$ population connections (e.g., the edge set) within a population graph. The statistical methodology for the determination of conditional independence is provided in Dyer and Nason (2004) and software is available from the authors web server.

Biologically, conditional genetic independence is created and maintained by evolutionary processes such as migration, selection, drift, and mutation. Pair-wise migration between populations removes genetic independence by

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homogenizing the within population genetic structure of each population. When populations freely exchange migrants, allele frequency changes in one population are reflected in quantitative changes in the structure of the migrants arriving at neighboring populations. This migrant-induced connectivity maintains genetic dependence amongst populations exchanging genes. Conversely, sets of non-interacting populations may exhibit conditional independence as the cessation of gene exchange allows the populations to proceed on independent evolutionary trajectories. This manuscript focuses on the following questions the evolution of intraspecific genetic structure within the population graph framework.

(1) To what extent are topological features within a population graph related to common population genetic parameters?
(2) How does the manner in which migrants are exchanged within a finite set of populations influence topological structure?
(3) Does the rate at which genes move among populations influence topological structure?

At this point, closed form solutions to these conditional genetic processes have yet to be derived. As a result, each of these questions is addressed by means of examining within and among population genetic structure within an extensive battery of individual-based Monte Carlo simulations.

## 2. The evolution of topological features

In this section, I examine the how graph-theoretic properties in population graphs evolve in unison with more traditional population genetic parameters. In particular, I focus on the creation and maintenance of genetic structure itself and the development of spatially relevant structure under a model of isolation by distance within a sample landscape.

### 2.1. The topology of genetic structure

A set of populations exhibit increasing genetic differentiation with increasing divergence of allele frequency spectra, up to a maximum value when all populations are fixed for at least two different alleles (e.g., Wright, 1978). Population divergence can result from a reduction in pair-wise exchange of genetic material, which also increases the probability of pair-wise conditional independence. As such, sets of populations who exchange genes will be more connected, topologically and genetically, than those subsets of populations who do not exchange genes. In general, the number of connections a node within a population graph has to other nodes is defined as its *degree* and I will denote the degree of the $i$th node as $d(v_i)$ taken in aggregate, a set of populations with high rates of migration will have low values of $\Phi_{ST}$ and many edges connecting them in a population graph, whereas populations with low rates of migration with have fewer edges and larger estimates of among population differentiation.

The extent to which a node is connected to other nodes in a graph is a general measure of that nodes *centrality* within the graph. For example, consider the two graphs depicted in Fig. 1, both of which contain five populations and four edges. In both of these example graphs, the population labeled 3, is the most *central* in the topology. One method that can be used to provide a quantitative measure of how central a particular node is within the topology is the degree of the node, which is defined as the number of edges incident upon the node. Nodes of high degree are, at least locally, well connected within the topology whereas nodes with few incident edges are less connected and thus less able to receive information percolating through the network. Population 3 in Fig. 1B is said to have a higher *degree centrality* than the same population in Fig. 1A simply because it has a larger local connectivity. For purposes of generality, degree centrality for any arbitrary node is standardized by the number of nodes in the graph for comparative purposes as

$$C_{D,v_i} = \frac{d(v_i)}{K-1}, \tag{1}$$

where $K = |V|$ is the number of nodes in the graph. In total, the average degree centrality for the entire graph is

$$\bar{C}_D = \frac{1}{K}\sum_{i=1}^{K} C_{D,v_i}. \tag{2}$$

In what follows, I will use average degree centrality, $\bar{C}_D$, as a surrogate measure for genetic structure. In doing this, I am going to disregard heterogeneity in centrality within a single topology so that I can focus solely upon the comparison among large numbers of replicate topologies. The characterization of within topology heterogeneity in centrality will be the focus of a subsequent manuscript.

### 2.2. The topology of isolation

For sets of populations whose patterns of dispersal are spatially limited, Slatkin (1993) showed that the genetic distances between populations will be correlated with the physical distances separating them, a condition referred to as isolation by distance. Specifically, the population parameter $F_{ST} \equiv (4Nm+1)^{-1}$ shows that the idealized

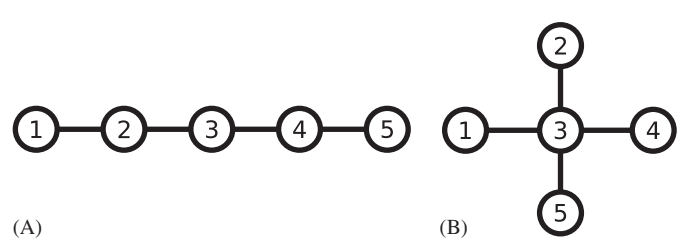


Fig. 1. Two example graphs that have the same sized node and edge set.

measure of among population variance in allele frequencies, $F_{ST}$ (and its numerous analogs), is inversely proportional to the product $Nm$, where $N$ is the genetic effective number of mating individuals and $m$ is the migration rate. From this, Slatkin (1981, 1985a, 1993) defined the statistic $\hat{M} \equiv Nm$ and showed that under dispersal models where dispersal is spatially limited relative to the distance separating pairs of populations, pair-wise estimates of $\hat{M}$ will be inversely proportional to physical distances (see also Rousset, 1997). Furthermore, the mode by which genes are exchanged between populations sets forth specific expectations for the magnitude of the coefficient of $\hat{M}$ regressed upon physical distance. For example, under Wright (1943, 1946) $N$-island model, the expectation of the regression coefficient is $E[\hat{M}_\beta] = 0.0$ as all populations are equally likely to mate with all other populations thereby negating the accumulation of any spatially relevant structuring. However, if populations are exchanging genes only with their most immediate neighbors, as in a one-dimensional stepping stone model (Kimura, 1953; Kimura and Weiss, 1964), the expectation for the regression coefficient is $E[\hat{M}_\beta] = -1.0$.

A set of populations exchanging migrants under either of these two idealized migration models also have significant implications for the distribution of edges within the topology. Under the $N$-island model, each pair of populations exchanges migrants with probability $1/K$. As a result, all populations are equally as likely to be connected to each other resulting in an overall topology that consists of nodes with individual high degree. Moreover, under this model of gene dispersal, there is no reason to expect any differences in the probability of two populations sharing a single edge, as all are sharing migrants with equal likelihood. In general, topologies that evolve under the $N$-island migration model are expected to have nodes of high degree and the topology is expected to be tightly compacted. Conversely, if populations are exchanging genes under a one-dimensional stepping stone model, populations that are not directly exchanging migrants have a reduced probability of being connected. Here nodes will have lower average degree, $\bar{C}_D$ and the overall topology will be elongated along the axis that corresponds to the one-dimension along which migrants are being exchanged. These expectations follow directly from the definition of conditional independence.

To quantify structural characteristics such as the compactness or elongation of the entire graph, we must examine the lengths of *paths* through the graph. For simplicity, consider a path from node $v_i$ to $v_j$, denoted as $p(v_i, v_j)$, as a sequence of node→edge→node connections that start at $v_i$ and end at $v_j$ possibly passing through one or more other nodes within the topology. Assuming that a path exists (e.g., the graph is connected) and that each node and edge is only visited once in the course of walking the path (e.g., no looping), the length is defined as the sum of the edge weights of all intervening edges connecting $v_i$ to $v_j$. For any two populations in a graph, $v_i$, $v_j$ $(i,j \in V,\ i \neq j)$, there may exist several alternate paths, at least one of which is guaranteed to be the shortest. For a completely connected graph with $K$ populations, there exist $K(K-1)/2$ shortest paths, one for each unique pair of populations. The pair of populations whose shortest path is the largest in magnitude within the entire graph defines the graphs *diameter*, $d_G = \max_{v_i,v_j} p(v_i, v_j)$, which is analogous to the major axis of the graph.

The concept of graph diameter is also demonstrated in Fig. 1. Assuming that each edge is of unit length, the shortest path with the largest magnitude in graph depicted in Fig. 1A is $p(1,5) = 4$. While this simplified graph only has a single shortest path for all pairs of populations, the elongation along the central axis is consistent with the expectation of a topology whose populations are exchanging migrants under the one-dimensional stepping stone model. The diameter of the graph in Fig. 1B has several paths of equal length: $p(1,2) = p(1,4) = p(1,5) = p(2,4) = p(2,5) = p(4,5) = 2$.

It should be noted that while the average degree of a graph, $\bar{C}_D$, influences the diameter, $d_G$, the diameter is ultimately determined by the order in which the populations are connected. From Eq. (2), the average degree from both graph are $d_{G,A} = d_{G,B} = \frac{8}{5}$. Therefore, diameter provides a complimentary measure of topological structure that is not directly available by inspection of the degree distribution alone. This notion is central to the subset of graph theory interested in small-world dynamics (Watts and Strogats, 1998; Strogatz, 2001).

### 2.3. The simulation model

A series of individual-based Markov chain Monte Carlo simulations were conducted following the general methods outlined in Slatkin (1981, 1985b) to quantify the simultaneous evolution of the two sets of paired parameters: (a) genetic structure ($\Phi_{ST}$) and degree centrality ($\bar{C}_D$), and (b) isolation by distance ($\hat{M}_\beta$) and graph diameter ($d_G$). Each simulation run consisted of $K = 25$ populations, which contained $N = 100$ diploid individuals. Generations were discrete, non-overlapping, and consisted of three ordered phases; mutation, mating, and migration. At the beginning of each simulation run, all populations were created de novo from randomly selected allele frequencies for ten independently assorting co-dominant loci each of which had three alleles. All alleles assorted independently and there was no linkage. An infinite alleles model was assumed and the occurrence of a mutation was modeled as a Poisson process with parameter $\mu = 10^{-5}$ constant across the simulations. Mating proceeded by selecting random pairs of individuals to act as parents for a single offspring. Population sizes were kept constant throughout the simulations and selfing was strictly disallowed.

To examine how the pattern of gene movement influences the evolution of genetic and topological structure, the simulations were run using both the $N$-island and one-dimensional stepping stone models following the

methods outlined above. During each replicate run, the multilocus genotypes were sampled at the beginning of each run and every five generations thereafter. For each sampled data set, the parameters $\Phi_{ST}$, $\bar{C}_D$, $\hat{M}_\beta$, and $d_G$ were estimated. The parameter $\hat{M}_\beta$ was estimated from the parameter $\theta$ using a modification of Slatkin's code (ibd.c, available from the author upon request). Estimation of the population graph for each sample followed the methods outlined in Dyer and Nason (2004). A fixed migration rate of $m = 0.10$ was set for all simulations and each run was allowed to proceed for 1000 generations. While this migration rate is somewhat large, it was sufficient to allow all populations to reach migration/mutation equilibrium during the course of the simulation runs. The consequences of alternate migration rates is explored in the following section. For each migration model, 100 replicate simulations were performed and the analytical 95% confidence intervals on the mean were ascertained. In total $1.0e^9$ genotypes were analyzed for this section.

In general, the overall topological differences between the $N$-island and stepping stone models are readily apparent when visualizing the topologies. The two topologies depicted in Fig. 2 represent the populations at the end of 1000 generations of mating (by which time equilibrium has been attained). The $N$-island model of migration results in an overall topology that is compact, containing nodes of high degree (Fig. 2A). Topologies resulting from mating following a stepping stone model were much more elongated, contained nodes of lower degree, and exhibited a primary axis corresponding to the one-dimension of the stepping stone model (Fig. 2B). The remainder of this section examines the quantitative differences in these topologies.

The simultaneous evolution of genetic structure, as measured by $\Phi_{ST}$ and average degree centrality, $\bar{C}_D$, for both the $N$-island and stepping stone models are presented in Fig. 3. Under the $N$-island model, $\Phi_{ST}$ quickly stabilizes and remains unchanged throughout the simulation runs

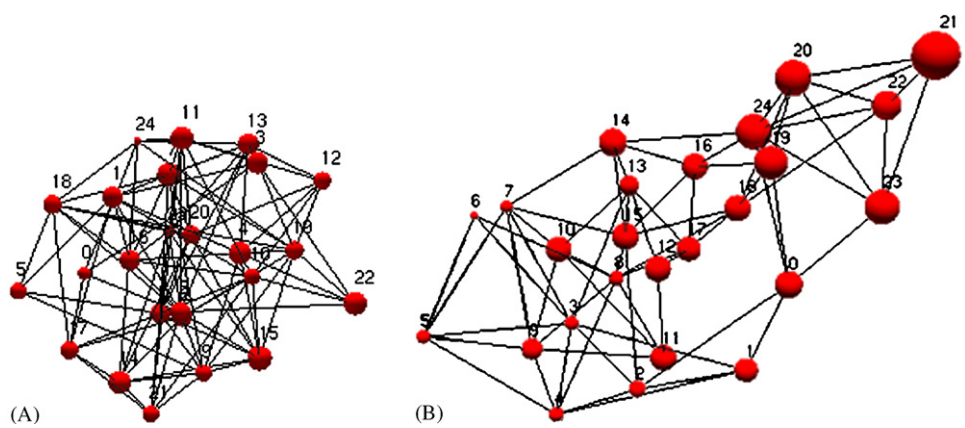


Fig. 2. Population graphs from the simulations depicting: (A) the $N$-island model of gene movement across the landscape and (B) gene movement following a one-dimensional stepping-stone model.

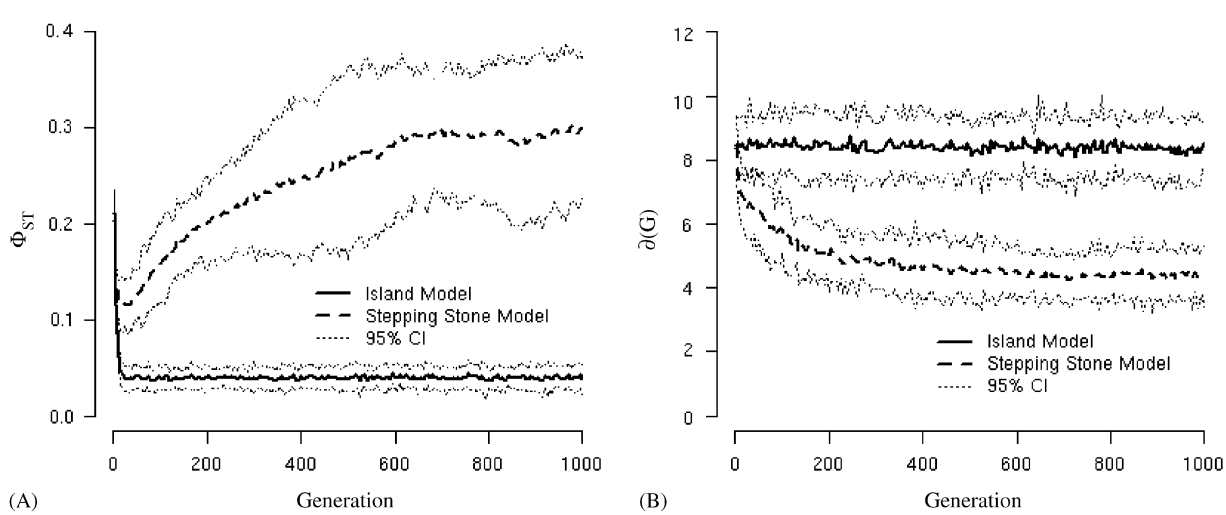


Fig. 3. The simultaneous evolution of among population genetic structure as measured by: (A) average among population variation, $\Phi_{ST}$ and (B) average node centrality, $\bar{C}_G$. Ninety-five percentile confidence intervals (dotted lines) surround the mean values for both the Island (solid line) and stepping stone (dashed line) models of migration.

(Fig. 3A solid line) as does $\bar{C}_D$ (Fig. 3B solid line). This behavior is expected as relatively few generations are required to reach equilibrium if all populations are mating at random with respect to strata. Under the stepping stone model, stabilization of the parameters takes considerably longer with $\Phi_{ST}$ stabilizing just after 800 generations and $\bar{d}_G$ stabilizing at around 500 generations. However, given a migration rate of $m = 0.10$, the time to stability is consistent with that reported by Slatkin (1985b). At equilibrium, $\Phi_{ST} = 0.038 \pm 0.008$ for the island model and $\Phi_{ST} = 0.298 \pm 0.036$ for the stepping stone model.

Under the stepping stone model of migration, the parameter $\Phi_{ST}$ initially declined until 20–30 generations at which time the trajectory changed and the parameter began to increase towards its final plateau (dashed line in Fig. 3A). This non-linear behavior in $\Phi_{ST}$ results primarily from the random initial conditions in the populations at the beginning of each simulation run that require a short *burn in* period to stabilize the among population component of genetic variation, $\sigma_A^2$. After $\Phi_{ST}$ begins the upward trajectory towards its final plateau, the correlation between $\Phi_{ST}$ and $\bar{C}_D$ is highly significant; $\rho = -0.950$ (Pearson product-moment; $P \leqslant 2.2e^{-16}$; $df = 178$; $t = -40.805$). These results suggest that the among population component of genetic variation and individual node connectivity both respond in a similar manner to population genetic processes.

The evolution of spatially explicit population genetic structure, as quantified by the regression coefficient for isolation by distance $\hat{M}_\beta$ and graph diameter $\bar{d}_G$, responded in a manner similar to that demonstrated for $\Phi_{ST}$ and $\bar{C}_D$ (Fig. 4). Under the island model, where populations randomly exchange genetic material, $\hat{M}_\beta = 0$, as expected. Similarly, the diameter of a population graph under this migration model remained small and unchanged throughout all simulations. Under the stepping stone model, $\hat{M}_\beta$ approaches and stabilizes on $-1.0$ around 1000 generations, as predicted in Slatkin (1993), and $\bar{d}_G$ reaches its equilibrium value of $\bar{d}_G = 18.1$ within 500 generations.

The correlation between the slope parameter, $\hat{M}_\beta$, and graph diameter, $\bar{d}_G$ showed a slightly larger correlation than noted for the evolving values of structure and centrality. Across the entire set of simulation runs, $\rho = -0.976$ (Pearson product-moment; $P \leqslant 2.2e^{-16}$; $df = 166$; $t = -57.1549$). Overall, these simulation results suggest that the joint evolution of within and among population genetic structure within a set of interacting populations can readily be examined from either a traditional population genetic perspective or from an analysis of topological features contained within a population graph.

## 3. The topological consequences of migration

In this section, I examine the consequences of variation in migration rate on the topological structure. While a migration rate of $m = 0.10$, as was used in the previous section, may be rather large for many natural populations, it was selected such that the simulation runs would attain equilibrium within 1000 generations and so that populations would be guaranteed to not become fixed for particular alleles. Here, I reduce the migration rate in the simulations to understand how the rate of gene movement influences the time to stability in population genetic and topological parameters. Furthermore, the reduction in gene movement results in a concomitant increase in the probability of fixation for individual populations and the behavior of topological change under fixation is also described.

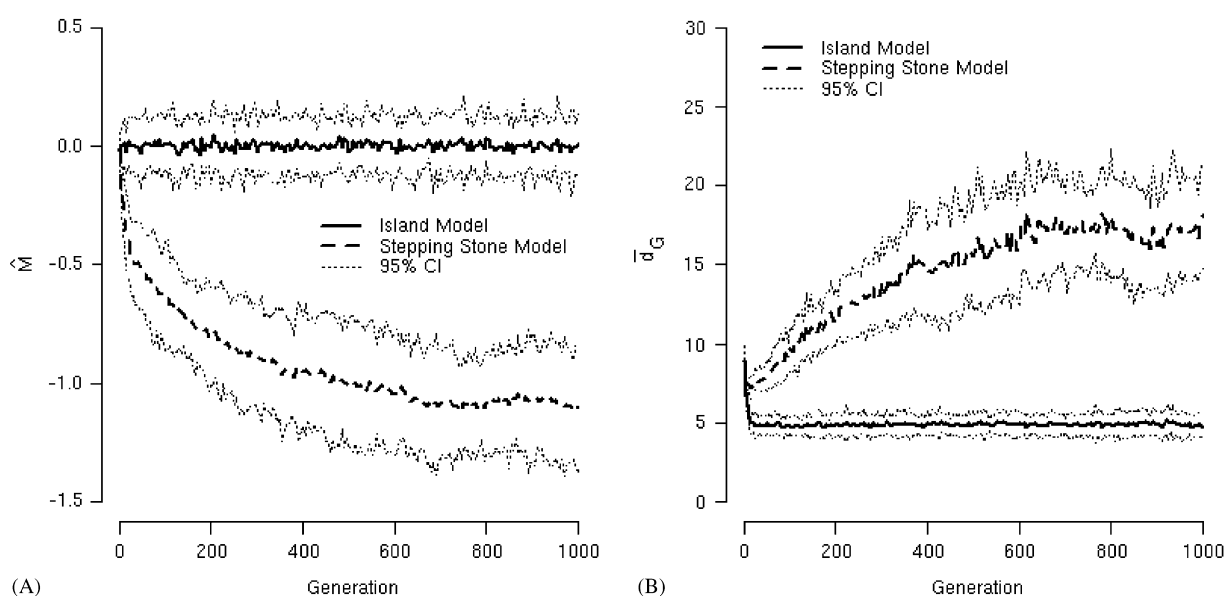


Fig. 4. The evolution of isolation by distance under both the $N$-island and stepping stone models as measured by: (A) Slatkin's $\hat{M}$ (from $\theta$) and (B) graph diameter.

For these simulations, I used a serial reduction in symmetric migration rates of $m = \{0.100, 0.010, 0.001\}$. Under these conditions, individual runs were extended to encompass 2000 generations to allow populations exchanging migrants are lower migration rates to achieve equilibrium. As before, populations were sampled at $t = 0$ and every five generations thereafter. All other parameters were held constant as before. Empirical 95% confidence intervals on the means presented below were determined from the analysis of $3.0e^{14}$ diploid genotypes.

Among population genetic structure, as quantified by the statistic $\Phi_{ST}$, increased with a reduction in pair-wise migration rates (Fig. 5A). The equilibrium value for $\Phi_{ST}$ when $m = 0.100$ was not significantly different than that estimated in the previous section. Furthermore, the trajectory reversing behavior of $\Phi_{ST}$ for $m = 0.10$ was not observed for lower migration rates. The most restrictive migration rate, $m = 0.001$, resulted in fixation (e.g., $\Phi_{ST} = 1.0$, for the sets of populations within each simulation run. The intermediate migration rate, $m = 0.010$, did not reach equilibrium during the course of the simulation runs and appeared to be steadily increasing towards fixation. Examination of the $\Phi_{ST}$ values between generations $1500 - 2000$, revealed a significantly non-zero slope (regression: $df = 1, 99$; $F = 1135$; adjusted $R^2 = 0.919$ $P < 2.2e^{-16}$), suggesting that the populations had yet to reach equilibrium. Assuming the trajectory from these values predicts that fixation would for these populations would be reached at generation 6145.

Degree centrality ($\bar{C}_D$), stabilized rather quickly for the largest migration rate (Fig. 5B). The evolution of $\bar{C}_D$ under the most restricted migration rate, $m = 0.001$, demonstrates clear non-linearity, driven in part by the process of fixation. Once a pair of populations became fixed for an allele, it is impossible for them to maintain conditional genetic dependence due to the lack of within population variation. As a result, individual populations, even if they are fixed for the same alleles, become detached from the graph forming isolated nodes with $C_{D,v_i} = 0$. As a result, $\bar{C}_D$, as averaged across replicate runs at the same generation, decreases rapidly until all populations are isolated and there are no edges in the graph (e.g., solid line Fig. 5B after generation 1000).

The intermediate migration rate took considerably longer to stabilize on a trajectory than the largest migration rate. As observed for the intermediate migration

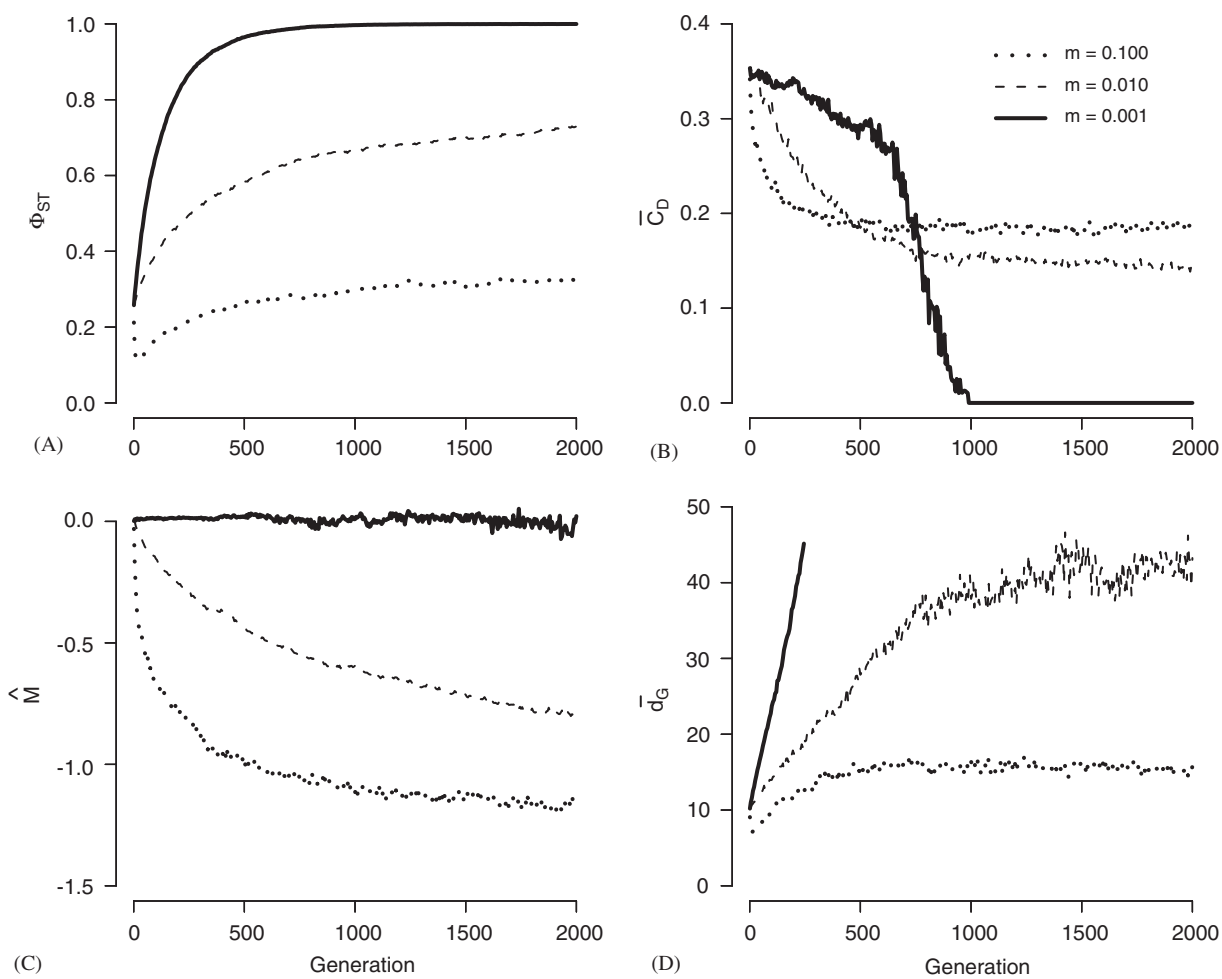


Fig. 5. The effects of alternate migration rates, $m = [0.001, 0.010, 0.100]$, under the stepping stone model on: (A) among population genetic structure, (B) node centrality, (C) isolation by distance, and (D) graph diameter. Empirical 95% confidence intervals not shown for clarity.

simulations of $\Phi_{ST}$, the slope coefficient of $\bar{C}_D$ during the last 500 generations was non-zero and declining (regression: $df = 1, 99$; $F = 87.11$; adjusted $R^2 = 0.4627$; $P = 3.118e^{-15}$). However, from the observations of $\bar{C}_D$ for $m = 0.001$, the decrease in $\bar{C}_D$ for the intermediate migration rates are not predicted to maintain a linear decrease through time and are expected to enter a non-linear phase once individual populations start achieving fixation as observed for $\bar{C}_D$ for $m = 0.001$. As a result, it is not clear as to how many generations would be required without further simulations.

The behavior of spatially explicit genetic structure, as quantified by the slope coefficient for isolation by distance ($\hat{M}_\beta$; Fig. 5C), was similar to that observed for $\Phi_{ST}$. As predicted by Slatkin (1993) and demonstrated in the previous section, the parameters for $m = 0.100$ under a one-dimensional stepping stone model stabilized quickly to $\hat{M}_\beta = -1$. For simulations under the most restricted migration rate, $\hat{M}_\beta = 0.0$, showing that the relative rate of migrants being exchanged along the one-dimensional landscape was sufficient low such that spatially relevant genetic structure was not observed. The intermediate migration rate, $m = 0.010$ appears to be approaching $\hat{M}_\beta = -1.0$ at the cessation of the simulation runs. However, it is predicted that this trajectory will reverse direction once loci within populations begin to become fixed. Following fixation, the relationship between pairwise differentiation and physical separation will be lost and $\hat{M}_\beta$ would immediately tend towards 0.0.

Finally, the response of graph diameter, $\bar{d}_G$, was similar to that observed for node centrality (Fig. 5D), albeit in the opposite direction. The most permissive migration rates resulted in a quick plateau as in the previous section. Intermediate migration rates resulted in topologies of increasing breadth. However, across the second half of the simulations, the behavior of $\bar{d}_G$ is not as apparent as that observed for node centrality.

For the most restricted migration rates, diameter grew very rapidly up to generation $t = 245$ at which point the diameter becomes undefined. This is in part due to the algorithm required to compute $\bar{d}_G$ as well as the definition of *path*. One of the assumptions in computing a path, $p(v_i, v_j)$, for any topology is that the graph is connected, thereby assuring a finite path length between any $v_i$ and $v_j$ ($v_i \neq v_j$; $v_i, v_j \in V$). For topologies that have at least one disconnected node, say $v_k$, $p(v_i, v_k) \equiv \infty$. As a result, out of all the shortest paths through a graph, the one whose length is $\infty$ is the largest, and thus by definition, the diameter. Across all simulations, the transformation from a single component to a set of disconnected sub-components appears to occur rather rapidly (Fig. 6). Relatively rapid phase transitions such as these are not uncommon features of dynamic graph topologies (see Bollobas, 2001, for an extensive discussion). At this time, there is not a clear understanding of how this phase transition takes place. It is possible to disregard disconnected nodes or subgraphs when the calculation of the diameter. However, to due so defeats the purpose of this exercise as one is then focusing on the diameters of increasingly small subgraphs as the entire topology evolves from completely connected to completely disconnected.

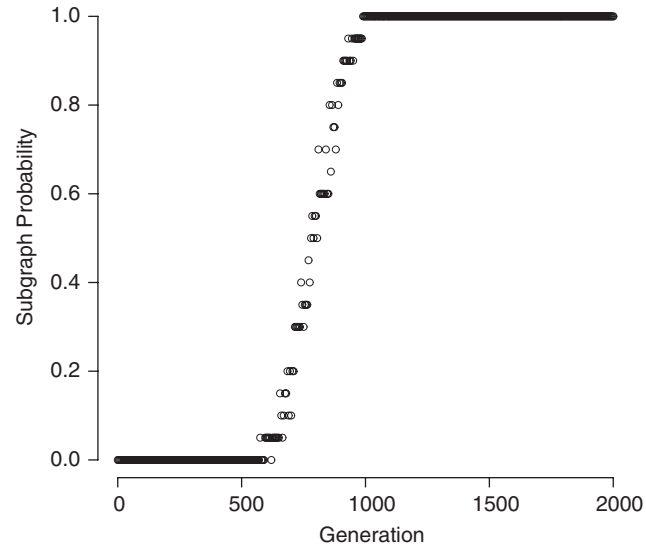


Fig. 6. The probability of observing a disconnected subgraphs within simulated data sets for the most restricted migration rate, $m = 0.001$. Disconnected components arise when insufficient gene movement allows populations to be genetically independent.

## 4. Discussion

It is not an accident that specific structural components within the topology of a population graph evolve in tight unison to more traditional population genetic parameters. The population graphs analysis arose from work in decomposing among population genetic variation for the purposes of developing post-hoc tests for the AMOVA framework. Because of this, a population graph can be viewed as an extension of the genetic variance decomposition in AMOVA albeit with a slightly different goal. AMOVA evaluates support for the null hypothesis $H_0 : \Phi_{ST} = 0$, which states that there is no structure among sampled stratum (Excoffier et al., 1992). The statistic $\Phi_{ST}$ is the ratio of the among stratum variance, $\sigma_A^2$, to the total genetic variance $\sigma_T^2 = \sigma_A^2 + \sigma_W^2$. If $\sigma_A^2$ is sufficiently large, given $\sigma_W^2$, then we can reject $H_O$. However, while the topology of a population graph, in aggregate form, contains both $\sigma_W^2$ and $\sigma_A^2$ and can be used to gain an estimate of the statistic $\Phi_{ST}$ (Dyer and Nason, 2004, Appendix A), this is not the ultimate goal of this approach. The decomposition of $\sigma_A^2$ into the components provided by each pair of strata defines a topology on the graph that reflects the operation of the evolutionary processes we are typically interested in describing. As a result, the analysis of topological features within a population graph demonstrates *how* variation is distributed amongst strata whereas the traditional AMOVA determines *if* there is a significant amount of variation present.

The *in silico* results presented introduce the concept of centrality and I opted to use simple node degree as a surrogate measure. In a population graph, two populations are connected by an edge if and only if they have a significant degree of genetic dependence, conditional on the remaining populations in the data set. This conditional covariance arises, for example, when populations exchange migrants. A set of populations that are exchanging migrants should therefore be tightly connected. Degree centrality captures the flavor of this kind of connectivity very nicely. However, it is important to recognize that degree centrality is just one estimator of centrality. One of the things that degree centrality does not capture is heterogeneous clustering across the entire graph. As pointed out above using the graphs in Fig. 1, despite the stark differences in overall topology, $\bar{C}_D$ is the same for both of these graphs. Moreover, if these graphs were actual population graphs rather than ad hoc examples, the estimate of $\Phi_{ST}$ would be identical for both graph topologies. Dyer and Nason (2004, Appendix A) used this 'robustness' of $\Phi_{ST}$ and other similar statistics as a primary motivation to adopt an approach that is more sensitive to *how* genetic variation is distributed.

Degree centrality and other associated measures such as closeness centrality (a measure of how far a population is from all others) and betweenness centrality (a measure of node importance based upon how many of the shortest paths the target node occupies) can play an important role is in the analysis of individual population placement. Dyer and Nason (2004) discussed the concept of relative importance of individual populations to the overall 'flow' of genetic information across a species' genetic topology. Behaviors of statistics that quantify the relative importance of individual populations, within the population graphs context, is an ongoing topic of research.

Isolation by distance is, by definition, intimately linked to genetic structure parameters such as $F_{ST}$. There is also an a priori expectation that node degree has a direct influence on overall graph diameter. In fact, for a given sized graph, $K = |V|$, the continued addition of edges to the graph is guaranteed to reduce overall diameter (e.g., Watts and Strogats, 1998; Bollobas and Riordan, 2004). Here I was concerned with overall assessment of isolation by distance and thus focused on the evolution of $\bar{d}_G$ and the slope coefficient from Slatkin's $\hat{M}$ statistic when regressed on physical distance. Under the $N$-island and stepping stone models, the slope coefficient, $\hat{M}_\beta$ is well characterized (Slatkin, 1993) and the comparison to overall graph diameter is illustrative of how the overall topology evolves as a set of populations achieves equilibrium. However, there are three important points that should be made regarding isolation by distance and graph diameter.

First, it should be noted that the statistic $d_G$, as depicted above, is an unscaled parameter that is dependent on the size of the graph. Here I was able to use it directly as a comparison because the simulations ensured that all graphs had the same number of nodes. For comparative purposes, when comparing diameters of several different graphs, $d_G$ should be standardized by an appropriate method such as dividing by the sum of all shortest paths (e.g., $\sigma_A^2$ in the AMOVA decomposition).

Second, the statistic $\hat{M}$ is estimated in a pair-wise manner by calculating a structure statistic, such as $F_{ST}$ or $\theta$, for each of the $K(K-1)/2$ possible pairs of populations. These parameters are estimated based solely upon individual pairs of populations taken in succession. Statistically, estimation of a population parameter such as $F_{ST}$ from a single pair of populations results in a much larger parameter variance than estimation of the same parameter from all strata combined simply due to sampling variance. It is doubtful, given the within stratum sample sizes typically observed for population genetic studies that much credence would be given to parameter estimates using only a pair of populations. In aggregate, these pair-wise estimates used for the identification of isolation by distance patterns are used with little regard to error variance and little oversight of their individual statistical properties. Conversely, path lengths through a graph are estimated from the topology resulting from the analysis of all populations simultaneously. As a result, the stability of isolation by distance calculations within a graph context are more robust due to the central limit theorem than those from a collection of pair-wise estimates.

Finally, the parameter $\hat{M}$ is expected to be inversely proportional and non-linear in physical distance. The parameter $\hat{M}$ typically exhibits statistical heteroscedasticity, or unequal variance across the range of physical distances. While more complete analysis is warranted, preliminary results from the analysis of thousands of simulated graphs suggest that path lengths are both linear in and proportional to physical distance and show no indications of in-equality in variance (data not shown); desirable properties for the analysis of spatially relevant genetic structure.

A striking result from the simulation work is the abruptness at which isolated components appear within the evolving topologies. This behavior suggests a relatively directed and predictable phase transition (Fig. 6). In general, phase transitions such as this are common components of graphical structures (Bollobas, 1985; Stauffer and Aharony, 1992). At limited symmetric migration rates all replicate topologies went from a completely connected topology to one with one or populations being genetically independent of all remaining populations with just a few hundred generations. Upon closer inspection, populations that become disconnected from the larger graph component are those that gone to fixation. For these populations, $\sigma_W^2 = 0.0$ thereby preventing any covariation with other populations (due to a zero in the denominator in the calculation of covariation). The earliest generation for which a population has gone to fixation is $t = 245$ (shown as end of solid line in Fig. 5D).

Alternate migration rates served primarily to attenuate the rate of approach to equilibrium conditions for both population genetic and graph theoretic parameters

examined herein. For higher rates of migration, the graph theoretic parameters appeared to reach equilibrium values faster than the population genetic parameters, although the differences are marginal at best when considering more restricted rates. It is important to point out that the goal here was not to exchange population genetic parameters with graph theoretic analogs, rather these simulation results show the relationship between topological features and more traditional population genetic parameters.

Finally, a key benefit of examining how intraspecific genetic variation is distributed within a graph theoretic context lies in the additional information that can be extracted from the topology itself. Visually, population graphs synthesize multivariate population genetic data in a manner that lends itself to visual interpretation. Moreover, graph topologies are structures that have a well defined body of mathematical and statistical properties that facilitate novel approaches to the analysis of population genetic data. Extensions to the population graph approach that are currently being investigated include determining heuristic methods that identify natural clusters of populations with the graph and the reconstruction of historical ordering in migration patterns.

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